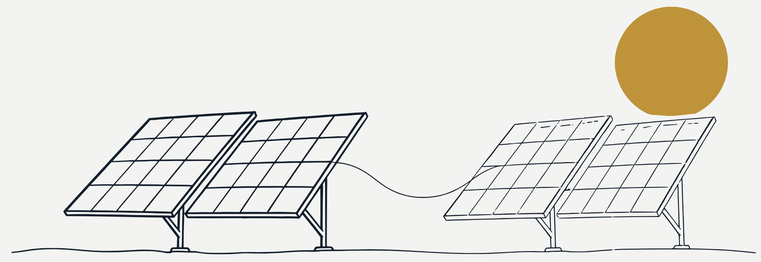
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SolarTwin AI

A trustworthy agentic digital twin for adaptive solar energy management



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Abstract

Forecasting models trained at one solar plant often lose accuracy at a new site. Our accepted NEECON 2026 paper measured both outcomes: successful transfer within a region and negative transfer across regions. **SolarTwin AI** is a software digital twin that checks transfer risk, compares forecast models, plans battery and flexible-load actions within hard limits, and records why each action was taken.

Problem Statement & Objectives

New plants have little generation history. A model borrowed from another plant can drift without warning, and an energy schedule built on that forecast inherits the error.

1. **Detect** when a transferred forecast should not be trusted.
2. **Decide** a safe battery and flexible-load schedule within operating limits.
3. **Explain** each decision with the evidence that produced it.

Related Literature

Solar forecasting usually depends on site-specific history. Domain adaptation (DANN, CDAN) aligns source and target features so a model can transfer without target labels; our ISGT Middle East 2025 study reached $R^2 = 0.75$ in blind cross-plant transfer [2]. Digital twins let operators test plant and storage decisions in simulation before acting on them.

Methodology

Data & scenarios
Plant telemetry, weather inputs and controlled shift events

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Forecast engine
CDAN + LSTM + Clear-Sky Index, Transformer, persistence

Trust monitor
Feature shift, model disagreement, risk flag

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Energy optimizer
Battery and flexible-load schedule under hard constraints (SciPy / OR-Tools)

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Explanation & audit
Decision trace linked to inputs, forecast, risk signals and constraints

Agents call the forecasting, monitoring and optimization tools. The language model only writes explanations; every number and constraint is computed in code.

Evaluation protocol

A Normal operation
Site matches the training domain.

B Sudden cloud event
Output drops; the battery plan adapts.

C Transfer-risk event
Seasonal mismatch triggers the fallback.

D Demand spike
Optimizer reports any unmet demand.

Metrics: MAE, RMSE and R^2 for forecasts; precision, recall and false-alarm rate for the risk monitor; operating cost and unmet demand for the energy plan.

Design & Validation

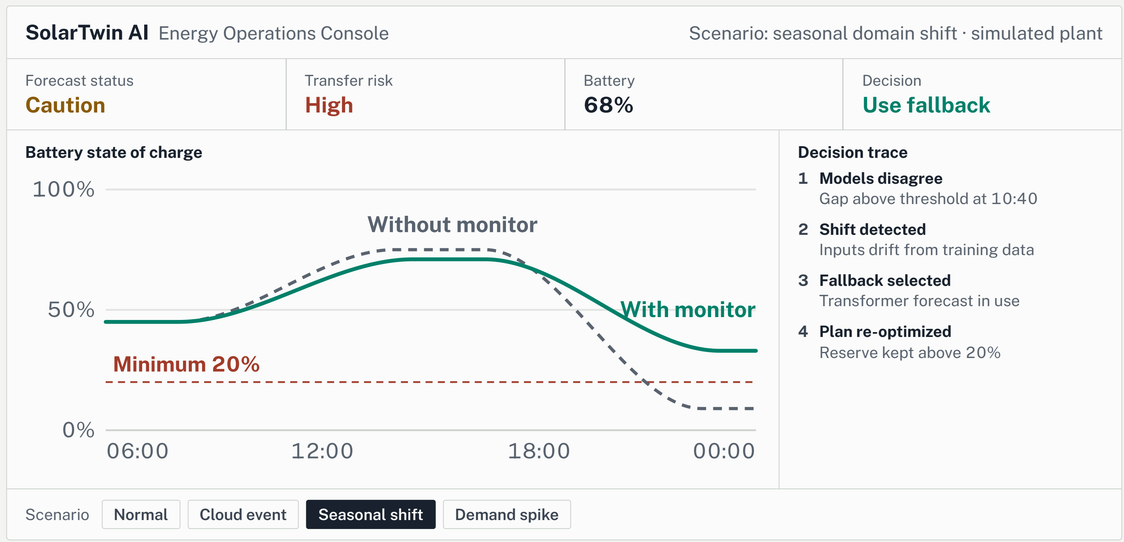


Fig. 1. Energy Operations Console. Judges pick a scenario and watch the forecast, risk signal and battery plan respond. Simulated plant and data.

Results & Analysis

Transfer case [1]	Model	Result
Plant 1 → Plant 2 Same region, capacity mismatch	CDAN + LSTM + CSI	**$R^2 = 0.8325$**
India → Australia Cross-regional	CDAN + LSTM + CSI	**Negative transfer**
India → Australia Cross-regional	Transformer, no adaptation	**Positive skill**

Adaptation helps when sites are similar and can hurt when they differ, so transfer risk must be measured before a forecast drives a decision. Fig. 2 shows how the monitor responds in a simulated shift.

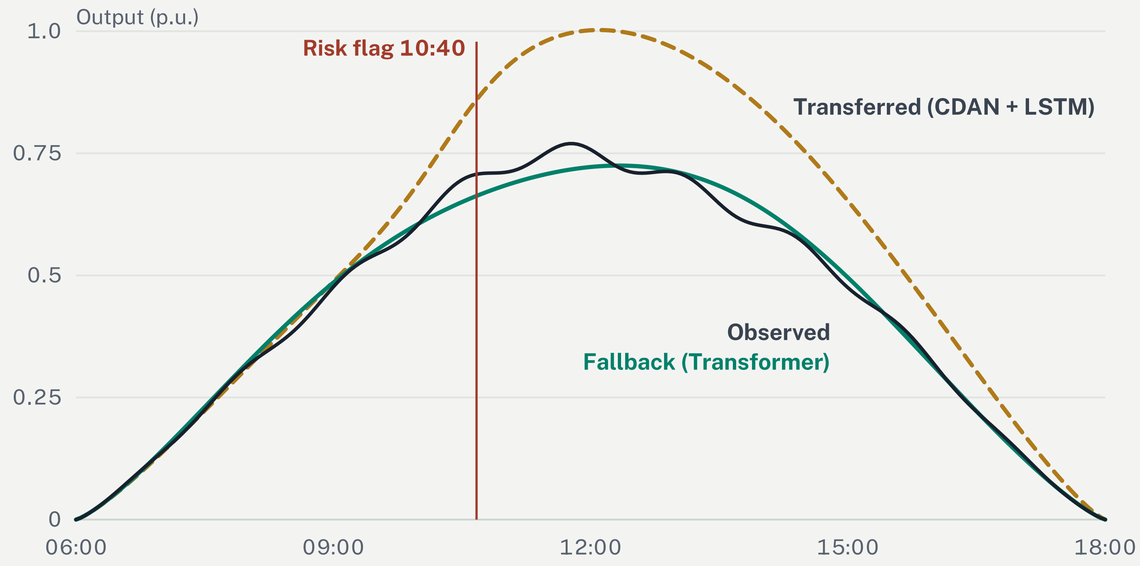


Fig. 2. Forecast candidates against observed output in a simulated seasonal shift. After the risk flag, the fallback forecast is used.

Conclusion

Takeaway	Forecast trust can be measured and acted on before it affects operations.
Theory	Links capacity-aware transfer learning with risk monitoring and constrained control.
Practice	A low-cost, offline tool to test new sites, weather events and demand changes.
Future work	Report detector precision, recall and false alarms; compare cost and unmet demand with baselines.

References & Contact

[1] Belay et al., “Learning When to Transfer: Robust Solar Power Forecasting Under Capacity Mismatch,” accepted at NEECON 2026.
[2] D. E. Belay et al., “Domain Adaptation for Cross-Plant Solar Power Forecasting Using Time Series Deep Learning Models,” IEEE PES ISGT Middle East 2025.